



SkillVane IT Academy
SKILLS THAT SHAPE CAREERS

2026 CURRICULUM

PROFESSIONAL CAREER PROGRAM

Multi-Cloud Data Engineer Program

(GCP + AZURE)

with Case Studies and Real-world Projects

STRUCTURED LEARNING • HANDS-ON LABS • INDUSTRY PROJECTS



COURSE TRAINER

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Course Snapshot

ROLE

Multi-Cloud Data Engineer

COURSE DURATION

4 months

MODE

Online

LANGUAGE

English

EVERYTHING AT A GLANCE

SkillVane IT Academy

Learn • Build • Advance

COURSE ROADMAP

Your Learning Journey

A structured path from cloud foundations to production-ready data engineering on GCP and Azure.

01 Cloud Introduction

What is Cloud Computing? • The need for cloud computing in modern businesses. • Why organizations moved to Cloud • Cloud Service Providers – GCP, Azure, AWS, ...

02 GCP Cloud Basics

GCP Cloud basics • GCP interfaces • GCP Locations • GCP Walkthrough • GCP IAM & Admin • Setting up a Free Account in GCP

03 Linux Basics on Cloud Shell

Getting started with Linux • Linux Installation • Basic Linux Commands • Cloud shell tips • File and Directory Operations • Text Processing

04 Python for Data Engineer

Data Types • Data Structures • File I/O operations • Exception Handling and Logging • REST API basics • Modules and Packages

05 GCP Data Engineering Tools

Google Cloud Storage • Cloud SQL • BigQuery (SQL development) • DataProc (Pyspark Development) • DataFlow (Apache Beam development) • Cloud Pub/Sub (Streaming) • Cloud Composer (DAG Development)

06 Azure Data Engineering Tools

Azure Introduction • Azure DataLake Storage (ADLS) • Azure SQL DB • Azure Data Factory (ADF) • Azure Synapse

07 Databricks on Azure/GCP

Lakehouse Platform • Unity Catalog • Delta Lake • ELT Pipelines • Case Study 1 • Case Study 2

08 By the End of the course What Students can Expect

Build end-to-end Data Engineering pipelines on GCP & Azure • SQL, Python, PySpark, Linux & Apache Beam • Medallion Architecture

Cloud Introduction

MODULE 01

Cloud Introduction

4 FOCUSED TOPICS

01 Cloud Computing

- What is Cloud Computing?
- The need for cloud computing in modern businesses.
- Why organizations moved to Cloud
- Cloud Service Providers – GCP, Azure, AWS, ...

GCP Cloud Basics

MODULE 02

GCP Cloud Basics

7 FOCUSED TOPICS

01 GCP Cloud basics

- Introduction to Google Cloud Platform (GCP) and its core services

02 GCP interfaces - Console, Cloud Shell and Cloud SDK

03 GCP Locations - Regions, Zones & Global Infrastructure

04 GCP Walkthrough - GCP Projects, Organizations, Folders & Billing Accounts

05 GCP IAM & Admin - Principals, Roles, Service Accounts & Resource Hierarchy

06 Setting up a Free Account in GCP

Linux Basics on Cloud Shell

MODULE 03 Linux Basics on Cloud Shell 8 FOCUSED TOPICS

01 Getting started with Linux

- Linux Installation
- Cloud shell tips
- File Content Manipulation (cat, less, head, tail, grep)
- User and Permission related (whoami, id, su, sudo, chmod, chown)
- Basic Linux Commands
- File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm, nano)
- Text Processing (awk, sed, cut, sort, uniq)

Python for Data Engineer

MODULE 04 Python for Data Engineer 1 FOCUSED TOPIC

01 Python Learning Path

- Getting Started with Python
- Data Types – Strings, Operators, Numbers (Int, Float), Booleans
- Python Programming Constructs - if, elif, else statements, for loops, while loops
- Exception Handling and Logging
- try and except blocks, else, finally, raise blocks
- Writing Logs to File
- Modular Programming in Python
- Classes
- What is API
- Status Codes, Request Headers, JSON Responses
- Modules and Packages
- Importing
- Install packages
- Introduction to Python and basic functions
- Data Structures – Lists, Tuples, Dictionaries, Sets
- File I/O operations - Read, write, append, rename, remove, copy, move
- what is exception handling, why it is needed
- Why Logging? And Logging Levels
- Data and Time operations - Working date and time, Formatting, Parsing, Time delta
- Functions & Lambda Functions
- REST API basics
- HTTP methods (GET, POST, PUT, DELETE)
- Using the requests Library, Reading API Responses
- What is module and packages
- Built-in modules

01 Google Cloud Storage

- Overview of Cloud Storage as a scalable and durable object storage service.
- Use cases for Cloud Storage, such as data backup, multimedia storage, and website content
- Creating and managing Cloud Storage buckets.
- Setting access controls and permissions for buckets and objects.
- Versioning and Object Versioning
- Monitoring and logging for Cloud Storage operations.
- Understanding buckets and objects in Cloud Storage.
- Labs: using console & CLI to do below
- Uploading and downloading objects to and from Cloud Storage.
- Data Transfer and Lifecycle Management
- Integration with Other GCP Services

02 Cloud SQL

- Introduction to Cloud SQL
- Configuring database settings, users, and access controls.
- Importing and exporting data in Cloud SQL.
- Integration with Other GCP Services
- Introduction to DMS
- Manual: Export and Import method
- Creating and Managing Cloud SQL Instances
- Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- Backups and High Availability
- Managing database user roles and permissions.
- End to End Database migration Project
- Automation: Cloud SQL DMS method

03 BigQuery (SQL development)

- Introduction to BigQuery
- Use cases for BigQuery in business intelligence and analytics.
- BigQuery Data Sources and File Formats
- Working with complex data types
- Data Integration and Export
- Exporting data from BigQuery to various formats.
- Configuring access controls and permissions in BigQuery.
- Views
- Authorized Views
- BigQuery Slots – on demand, flat-rate, flex-slots
- Case Study-2: Enterprise Social Media analytics platform
- BigQuery Architecture
- Various method of creating table in BigQuery
- Native table and External Tables
- Working json data, nested, repeated and array data
- Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
- Real-time data streaming into BigQuery.
- BigQuery Views:
- Materialized Views
- Optimization techniques in BigQuery
- Case Study-1: implement a real-world analytics data platform for Spotify

04 DataProc (Pyspark Development)

- Introduction to Hadoop and Apache Spark
- What is Spark and Pyspark.
- Overview of DataProc as a fully managed Apache Spark and Hadoop service.
- Creating and managing DataProc clusters.
- Preemptible instances and cost optimization.
- How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- Submitting and monitoring Spark and Hadoop jobs on DataProc.
- Job debugging and troubleshooting.
- Case study-2: Processing real-time patient health data
- Understanding the difference between Spark and MapReduce
- Understanding Spark framework and its functionalities
- Cluster Creation and Configuration
- Configuring cluster properties for performance and scalability.
- Learning Pyspark:
 - How to perform multiple transformations
- Running Jobs on DataProc
- Use of initialization actions and custom scripts.
- Case study-1: Data Cleaning of Employee Travel Records
- Case study-3: Creating a pyspark job to support ML model creations

05 DataFlow (Apache Beam development)

- Introduction to DataFlow
- Understanding the difference between Apache Spark and Apache Beam
- Learning Apache Beam
- How to perform multiple transformations
- Case study-1: Template method of creating pipelines
- Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS
- Use cases for DataFlow in real-time analytics and ETL.
- How Dataflow is different from Dataproc
- How to read from multiple data sources – csv, text, json, parquet, database table, BigQuery tables
- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
- Case study-2: E-commerce Transaction Processing

06 Cloud Pub/Sub (Streaming)

- Introduction to Pub/Sub
- Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.
- Using the GCP Console to create Pub/Sub topics and subscriptions.
- Publishing and Consuming Messages
- Implementing subscribers to consume and process messages from subscriptions.
- Understanding the role of Pub/Sub in event-driven architectures.
- Creating and Managing Topics and Subscriptions
- Configuring message retention policies and acknowledgment settings.
- Writing and deploying code to publish messages to a topic.
- Case study-1: Streaming use-case using Dataflow

07 Cloud Composer (DAG Development)

- Introduction to Composer/Airflow
- Use cases for Composer in managing and scheduling workflows.
- Creating and configuring Composer environments.
- Monitoring and managing workflow executions.
- Orchestrating workflows involving BigQuery, DataFlow, and other services.
- Integrating with external systems and APIs.
- Handling errors and retries in Composer workflows.
- Logging and monitoring for Composer workflows.
- Level-2-DAG: Orchestrating the DataProc pipelines
- Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub
- Overview of Airflow Architecture
- Creating and Managing Workflows
- Defining and scheduling workflows using Apache Airflow.
- Integration with Data Engineering Services
- Coordinating ETL processes with Composer.
- Error Handling and Troubleshooting
- Debugging and troubleshooting failed workflow executions.
- Level-1-DAG: Orchestrating the BigQuery pipelines
- Level-3-DAG: Orchestrating the Dataflow pipelines

01 Azure Introduction

- Introduction to Azure Cloud Platform and its core services
- Azure Locations - Regions, Zones & Global Infrastructure
- Azure IAM & Admin - Identities, Roles, Service Accounts & Resource Hierarchy
- Azure interfaces - Console, Cloud Shell and Cloud SDK
- Azure Walkthrough – Subscriptions, Resources, Resource group & Management group
- Setting up a Free Account in Azure Cloud

02 Azure DataLake Storage (ADLS)

- Introduction to Azure Data Lake Storage (ADLS)
- Blobs, Containers, and Storage Accounts
- Why Use Azure Data Lake Storage?
- Integration with Analytics Services
- Data Management via Console, SDK, CLI, REST API
- Pricing Overview
- Cost Factors
- Data Partitioning
- Monitoring and Usage Tracking
- Managing ADLS via Console
- Programmatic Access via SDKs
- Core Components of ADLS Gen2
- Hierarchical Namespace
- Scalability, Cost-Effectiveness, Security
- Access and Management of ADLS
- Role-Based Access Control (RBAC) and Access Control Lists (ACLs)
- Pricing Tiers (Hot, Cool, Archive)
- Best Practices for ADLS
- Tiering Strategy for Cost Optimization
- Interactive Access Methods
- CLI/Shell Interaction

03 Azure SQL DB

- Introduction to Azure SQL Database
- Key Features
- Creating a SQL db for practice

04 Azure Data Factory (ADF)

- What is ADF and Use cases
- Integration runtime
- Datasets
- Activities
- Key Features of Azure Data Factory
- Creating and Configuring Pipelines
- Monitoring and Managing Pipeline Runs
- Multiple examples with ADF activities
- Best Practices for ADF
- Data Partitioning and Parallelism
- Core components of ADF and understanding & configuring
- Linked Services
- Pipelines
- Triggers
- Pipelines in ADF
- Activities and Triggers
- Orchestration
- Case study-1: End to End Incremental pipeline with ADF, ADLS, SQL DB.
- Designing Efficient Pipelines
- Monitoring and Optimization for Performance

05 Azure Synapse

- Synapse Workspace, SQL Pools & Architecture
- Dedicated & Serverless SQL concepts
- Data ingestion using Synapse Pipelines
- External Tables, Views & Data Lake integration
- Query performance optimization, partitioning & workload management

01 Databricks Lakehouse Platform

- Introduction to Databricks Lakehouse Platform
- Explore the architecture, including clusters, jobs, and the workspace environment.
- Setting Up and Administering a Databricks Workspace
- Admin tasks including user management, permissions, and billing.
- Collaborative data science using Databricks notebooks.
- Databricks Unity Catalog
- Organizing data assets and enforcing security policies.
- Understanding Delta Lake's ACID transactions and versioning capabilities.
- Creating Delta Lake Tables
- Building ELT Pipelines with Spark SQL and Python
- Case Study 1: Real-time data processing with Spark Structured streaming
- Performance Optimization in Databricks
- Databricks Architecture and Core Components
- Key components like Databricks Runtime, Spark, and Delta Lake.
- Step-by-step guide on workspace creation and management.
- Working with Notebooks and Clusters
- Cluster setup, management, and autoscaling.
- Unified governance solution for managing data and metadata across the lakehouse.
- Managing Data with Delta Lake
- Best practices for efficient data management.
- Querying and managing data across multiple file formats.
- Incremental Data Processing with Delta Lake
- Case Study 2: Databricks Autoloader for file-based ingestion utility

By the End of the course What Students can Expect

MODULE 08

Outcomes

8 FOCUSED TOPICS

- 01 Build end-to-end Data Engineering pipelines on GCP & Azure
- 02 Develop strong skills in SQL, Python, PySpark, Linux & Apache Beam
- 03 Work with industry standard data engineering services
- 04 Implement Batch & Real-Time Data Processing pipelines
- 05 Design Bronze → Silver → Gold / Medallion Architecture
- 06 Build production-ready projects and troubleshoot real-world data engineering scenarios
- 07 Understand Cloud, Data Lake, Data Warehouse & Lakehouse architectures
- 08 Prepare for Data Engineer interviews with practical, scenario-based knowledge

Thank You.

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